

40. The ^1H NMR spectrum of naphthalene shows two multiplets (Figure 15-16). The upfield absorption ($\delta = 7.49 \text{ ppm}$) is due to the hydrogens at C2, C3, C6, and C7, and the downfield multiplet ($\delta = 7.86 \text{ ppm}$) is due to the hydrogens at C1, C4, C5, and C8. Explain why one set of hydrogens is deshielded more than the other.

- 49.** The NMR and IR spectra for four unknown compounds A through D are presented below. Possible empirical formulas for them (not in any